

The impact of financial leverage on cost of capital and firm valuation and the determinants of the optimal capital structure using the cost of capital approach for the firms under the construction, infrastructure & allied services sub-sector of the Philippine Stock Exchange for the period 2001 – 2009

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Introduction

Statement of the Problem

What is the impact of financial leverage on cost of capital and market value of firm and what is the optimal capital structure of the selected firms under the construction, infrastructure & allied services sub-sector of the Philippine Stock Exchange?

Introduction

Objectives of the Study

- Financial leverage → cost of capital
- Financial leverage → P/E ratio
- Financial leverage → market value of the firm
- Optimal capital structure

Introduction

Null Hypothesis (H_{o1}): There is no correlation between degree of financial leverage and cost of capital.

Alternative Hypothesis: (H_{a1}): There is a significant correlation between degree of financial leverage and cost of capital

Null Hypothesis (H_{o2}): There is no correlation between degree of financial leverage and the price earning ratio.

Alternative Hypothesis: (H_{a2}): There is a significant correlation between degree of financial leverage and the price earning ratio.

Introduction

Null Hypothesis (H_{o3}): There is no correlation between degree of financial leverage and valuation of the firm.

Alternative Hypothesis: (H_{a3}): There is a significant correlation between degree of financial leverage and valuation of the firm.

Null Hypothesis (H_{o4}): Financial leverage does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a4}): Financial leverage statistically influences cost of capital.

Introduction

Null Hypothesis (H_{o5}): Firm size does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a5}): Firm size statistically influences cost of capital.

Null Hypothesis (H_{o6}): The firm's systematic risk does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a6}): The firm's systematic risk statistically influences cost of capital.

Introduction

Null Hypothesis (H_{o7}): Market capitalization does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a7}): Market capitalization statistically influences cost of capital.

Null Hypothesis (H_{o8}): Price earnings ratio does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a8}): Price earnings ratio statistically influences cost of capital.

Introduction

Null Hypothesis (H_{o9}): Asset turnover does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a9}): Asset turnover statistically influences cost of capital.

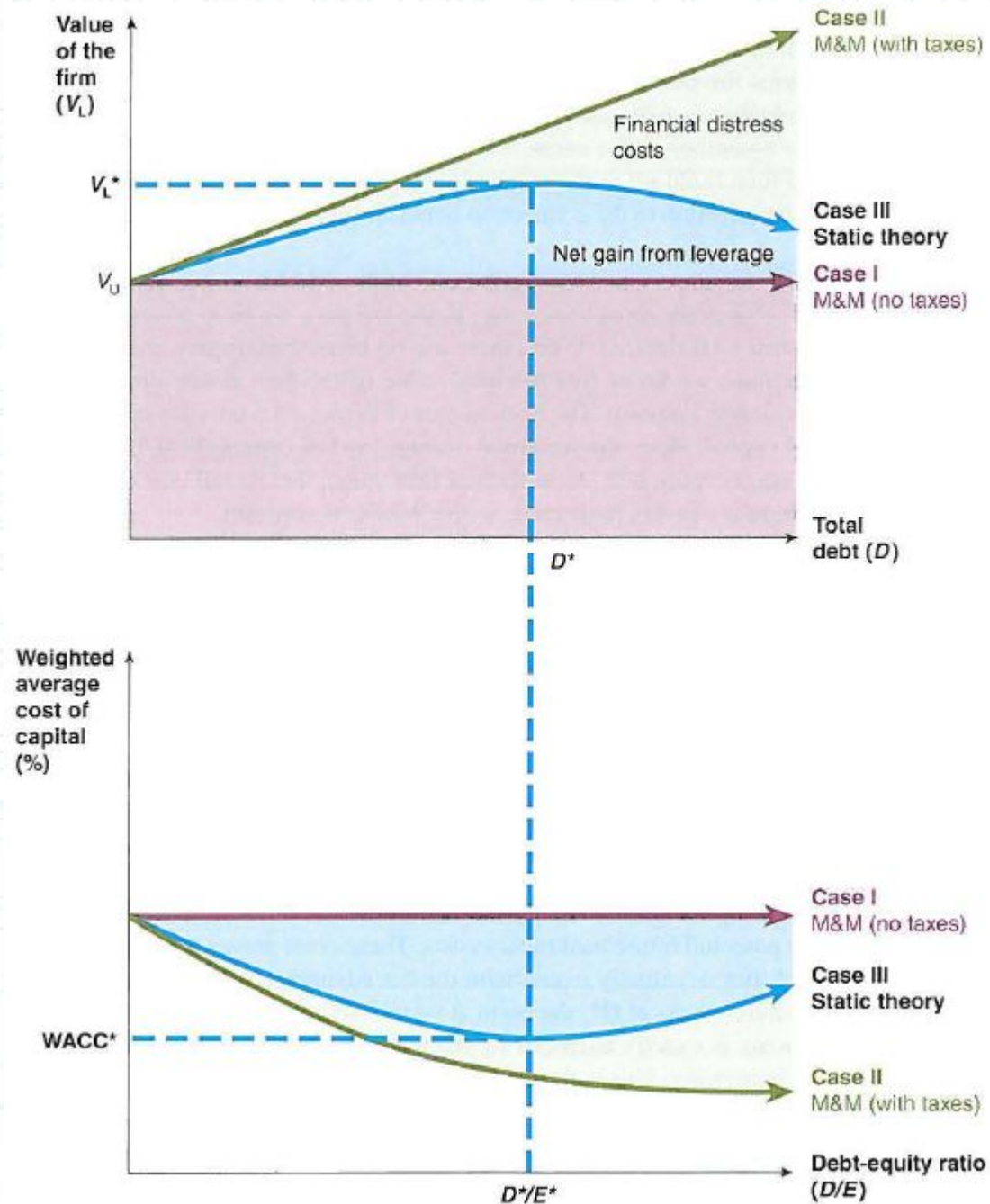
Introduction

Significance of the Study

- Financial Leverage
- Optimal Capital Structure
- Management's capital structure decisions

Framework of the Study

Static Tradeoff Theory



Framework of the Study

M&M Theorem

1st proposition – capital structure doesn't affect the cost of capital

2nd proposition – (+) debt, (+) leverage and cost of equity, but WACC stay the same

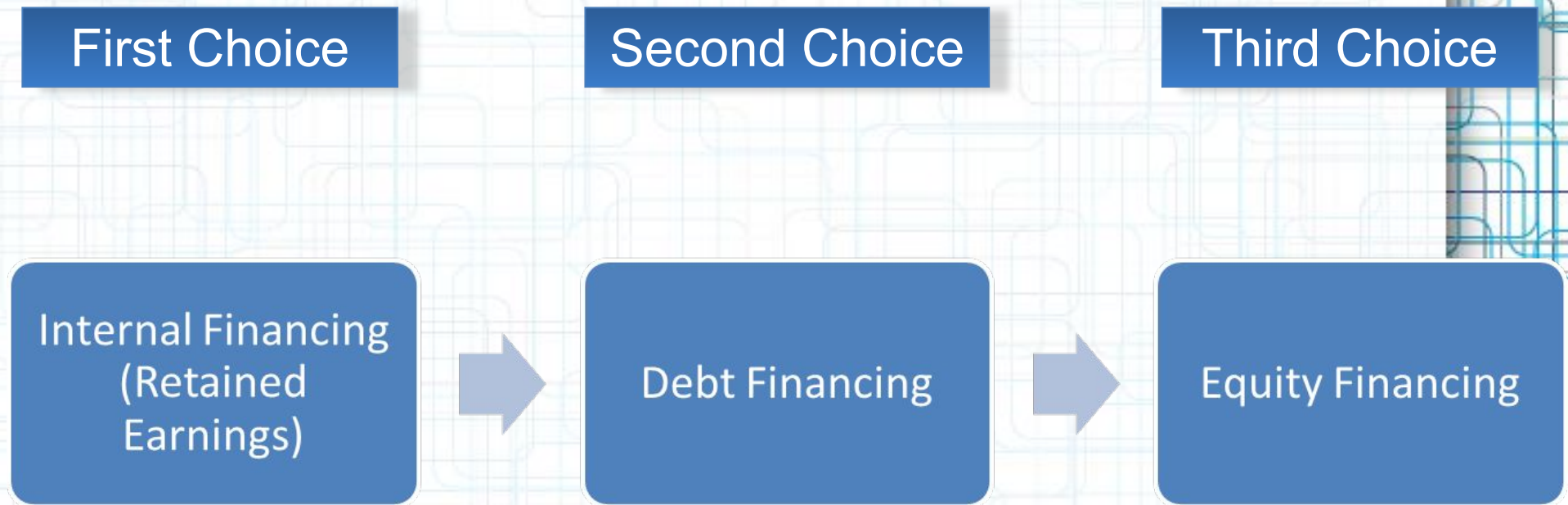
WITH TAX

1st proposition – (+) firm value as debt (+) due to tax shield

2nd proposition – when debt is used 100%, WACC drops where there is optimal capital structure

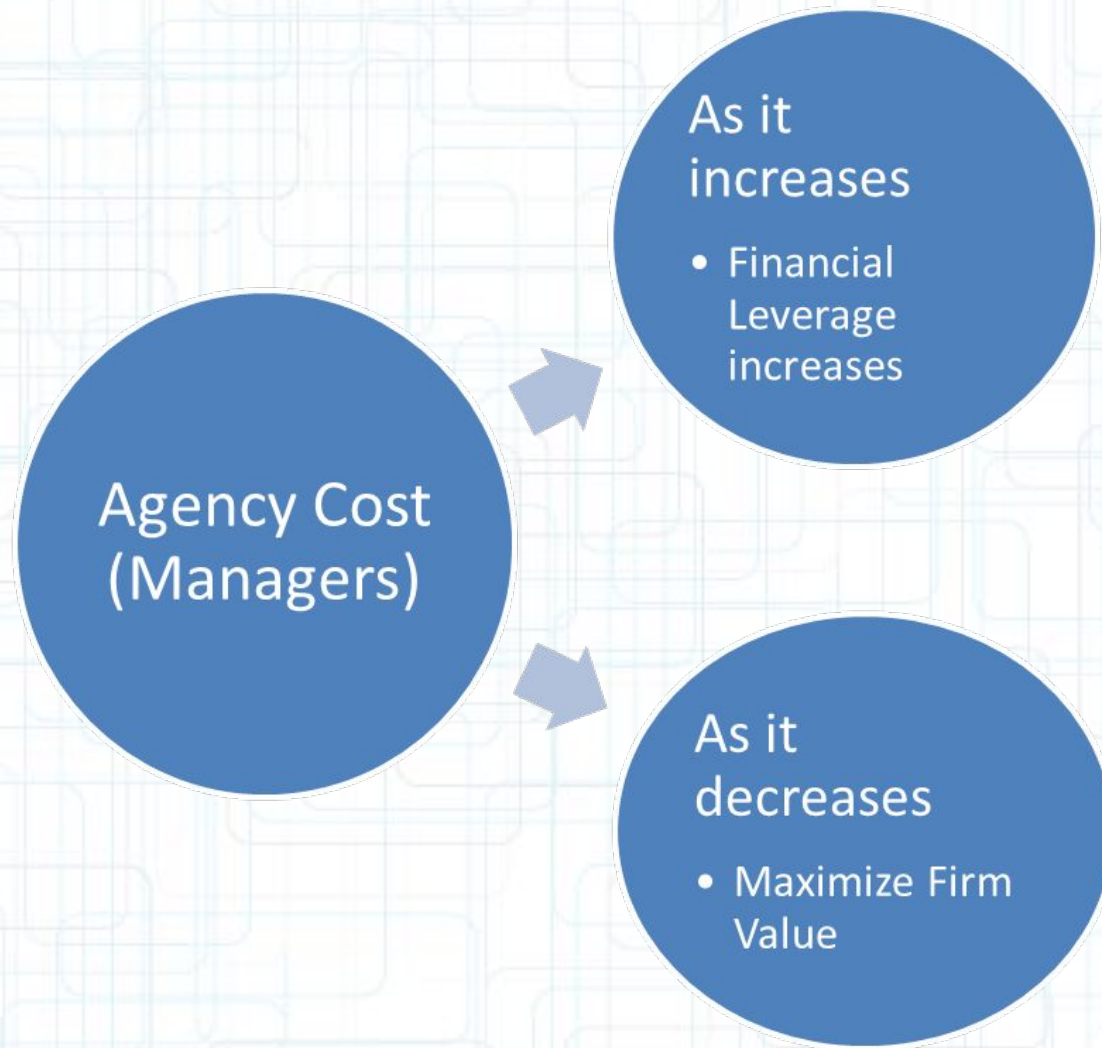
Framework of the Study

Pecking Order Theory



Framework of the Study

Agency Cost Theory



Methodology

Research Design

- Construction, infrastructure and allied services – publicly listed in the PSE
- Determining the relationship and significance of the dependent and independent variables
- Dependent variable – wacc
- Independent variables – financial leverage, squared financial leverage, price earning ratio, market capitalization, log of revenue, Asset Turnover, and Beta

Methodology

Research Design

- Formulas:
- Financial Leverage = Total Assets / Total Equity
- Weighted Average Cost of Capital (WACC) = (net worth / total assets) * K_e + (total external liabilities / total assets) * K_d of capital.
- Market Capitalization = share price * outstanding shares
- Price Earning Ratio = Price per share / earnings per share
- Asset Turnover = revenue / total sales
- Beta

Methodology

Sampling Design

- Sample size – 7
- Construction, infrastructure and allied services – publicly listed in the PSE.
- EEl Corporation, Holcim Philippines, Inc., Phinma Corporation, Republic Cement Corporation, Southeast Asia Cement Holdings, Inc., TKC Steel Corporation, and Vulcan Industrial & Mining

Methodology

Data Description and Collection Method

- Secondary data
- BLOOMBERG
- No. of years – 9 (from 2001-2009)

Methodology

Data Analysis Methods

- Gretl – Gnu Regression, Econometrics, Time-Series Library Software
- Stata - general-purpose statistical software package
- Panel data – different companies
- Karl Pearson's coefficient of correlation
- Multiple Regression and 2SLS
- T- test

Methodology

Test of Assumptions

- Multicollinearity – Variance Inflation Factor
- Heteroscedasticity – White Test

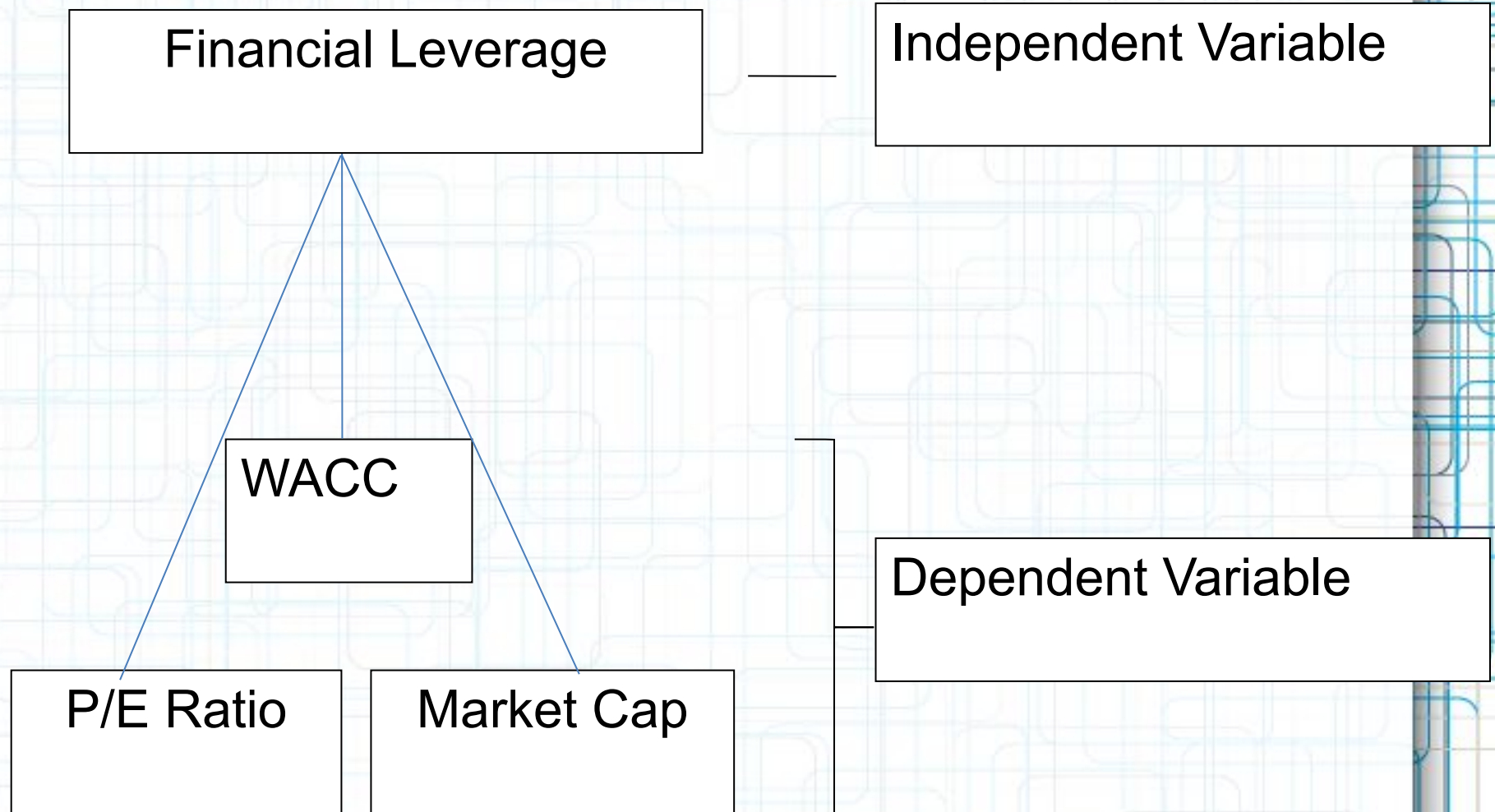
Methodology

Methodological Limitations

- Sampling limitations
- Secondary data reliability
- Remedy

Framework of the Study

Pearson's Coefficient of Correlation Diagram



Analysis and Interpretation of Results

Karl Pearson's Coefficient of Correlation

- Industry Coefficient Correlation

fin lev – market cap (-) p value = 0.1525

fin lev – pe ratio (+) p value = 0.9580

fin lev – wacc (-) p value = 0.1681

* all are insignificant, no correlation were present

Analysis and Interpretation of Results

Karl Pearson's Coefficient of Correlation

- High Levered Firms

fin lev – market cap (-) p value = 0.2776

fin lev – pe ratio (-) p value = 0.4296

fin lev – wacc (-) p value = 0.7607

* all are insignificant, no correlation were present

Analysis and Interpretation of Results

Karl Pearson's Coefficient of Correlation

- Low Levered Firms

fin lev – market cap (-) p value = 0.7664

fin lev – pe ratio (-) p value = 0.5076

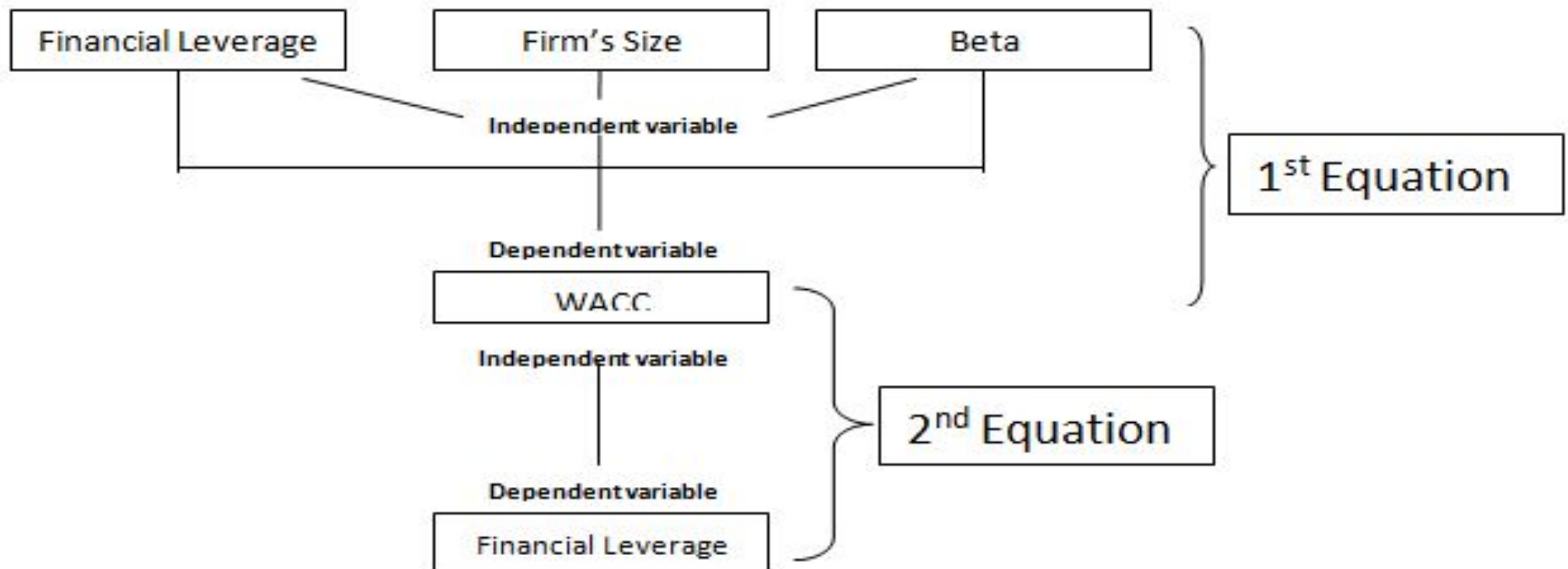
fin lev – wacc (-) p value = 0.5615

* all are insignificant, no correlation were present

Framework of the Study

Two Stage Least Squares Diagram

Figure 3.2



Analysis and Interpretation of Results

Two-stage Least Squares

Instrumental variables (2SLS) regression

Source	SS	df	MS	Number of obs = 63		
-----+-----				F(4, 58) = 2.03		
Model	-.539064448	4	-.134766112	Prob > F = 0.1026		
Residual	.626635876	58	.010804067	R-squared = .		
-----+-----				Adj R-squared = .		
Total	.087571428	62	.001412442	Root MSE = .10394		

wacc	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+-----						
finlev	-.2480918	.0873406	-2.84	0.006	-.4229231	-.0732605
finlev2	.026867	.0096002	2.80	0.007	.0076501	.0460839
lnrev	.0128323	.0082657	1.55	0.126	-.0037133	.029378
beta	-.0182334	.0221614	-0.82	0.414	-.0625944	.0261275
_cons	.2084395	.1544133	1.35	0.182	-.1006522	.5175313

Instrumented: finlev						
Instruments: finlev2 lnrev beta wacc						

Analysis and Interpretation of Results

Two-stage Least Squares

$$Wacc_{i,t} = 0.2084395$$

$$- 0.2480918(\text{fin_lev})$$

$$+ 0.026867(\text{fin_lev_2})$$

$$+ 0.01283323(\text{l_revenue})$$

$$- 0.0182334(\text{beta})$$

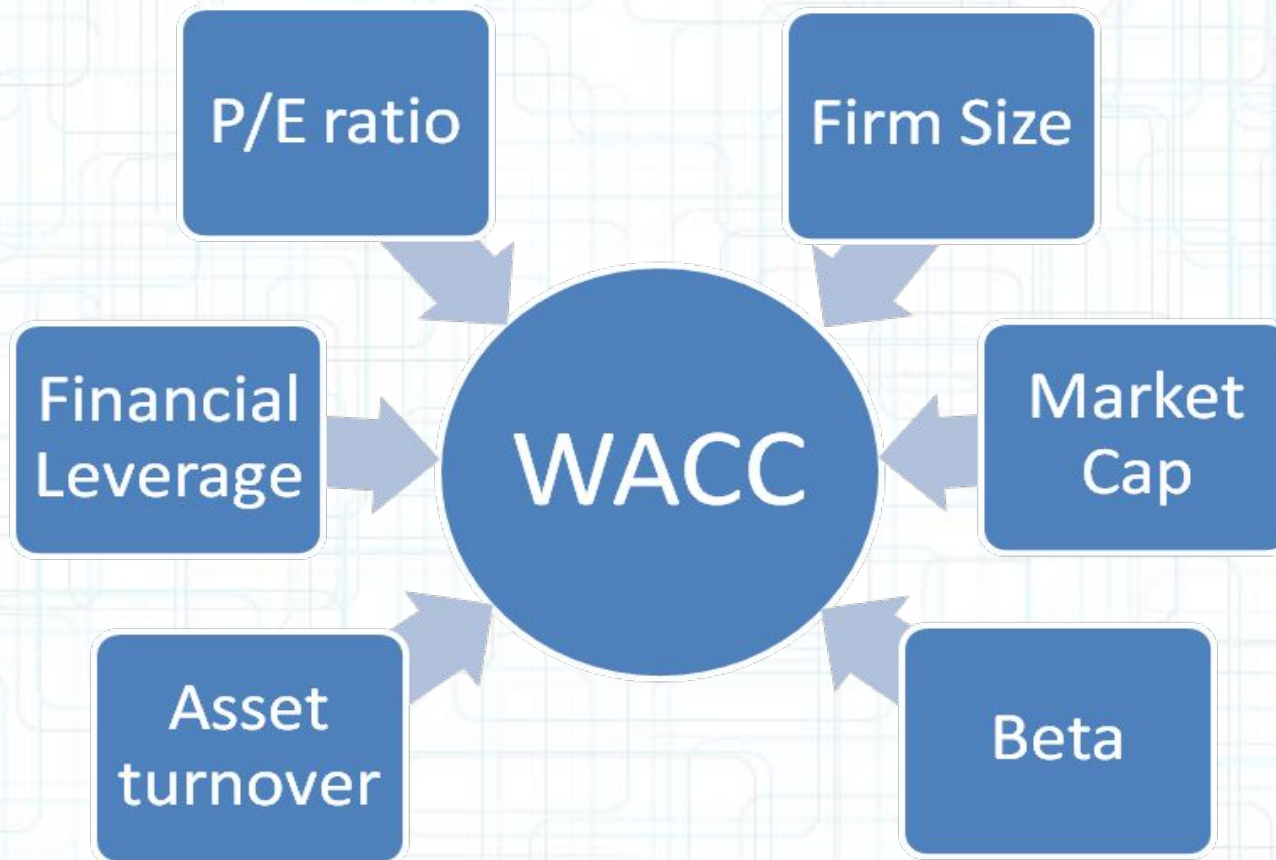
Optimal Capital Structure: 3.617

Analysis and Interpretation of Results

Difference of D/E leverage to optimal leverage			
	D/E	Optimal debt level	Difference
Cmt	1.1600	3.617	-2.4570
Eei	1.5100	3.617	-2.1070
Hlcm	0.3900	3.617	-3.2270
Phn	0.3700	3.617	-3.2470
Rcm	0.5600	3.617	-3.0570
T	1.2600	3.617	-2.3570
Vul	0.6900	3.617	-2.9270

Framework of the Study

OLS Method Diagram



Analysis and Interpretation of Results

Ordinary Least Squares

Model 12: Pooled OLS, using 63 observations

Included 7 cross-sectional units

Time-series length = 9

Dependent variable: wacc

	coefficient	std. error	t-ratio	p-value	

const	0.0905416	0.0434111	2.086	0.0417	**
l_revenue	0.00180284	0.00222927	0.8087	0.4222	
fin_lev	-0.0170925	0.00867615	-1.970	0.0539	*
fin_lev__2	0.00172614	0.000957317	1.803	0.0769	*
asset_turn	-0.0160653	0.00240008	-6.694	1.20e-08	***
pe_rat	-6.04725e-05	8.62642e-05	-0.7010	0.4862	
mkt_cap	-6.02543e-07	4.02699e-07	-1.496	0.1403	
beta	-0.00951100	0.00566452	-1.679	0.0988	*

Analysis and Interpretation of Results

Ordinary Least Squares

Mean dependent var 0.078443 S.D. dependent var 0.037258

Sum squared resid 0.038932 S.E. of regression 0.026605

R-squared 0.547649 Adjusted R-squared 0.490077

F(7, 55) 9.512417 P-value(F) 1.11e-07

Log-likelihood 143.3630 Akaike criterion -270.7260

Schwarz criterion -253.5810 Hannan-Quinn -263.9828

rho 0.104174 Durbin-Watson 1.546647

Excluding the constant, p-value was highest for variable 5 (pe_rat)

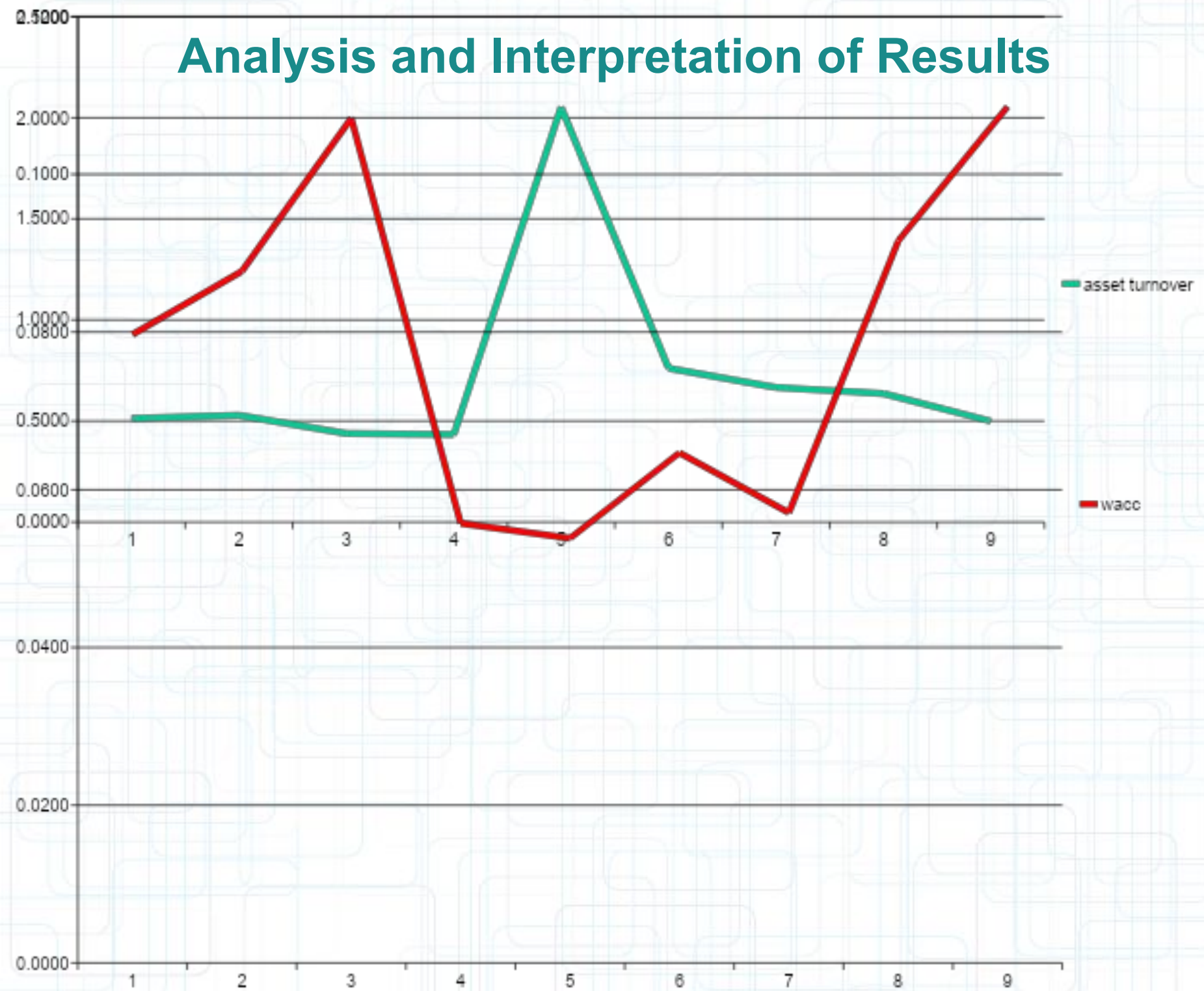
Analysis and Interpretation of Results

Ordinary Least Squares

$$Wacc_{i,t} = 0.0905416$$

- + 0.00180284 (l_revenue)
- 0.01709250 (fin_lev)
- + 0.00172614 (fin_lev__2)
- + 0.0160653 (asset_turn)
- 6.04725e-05 (pe_rat)
- 6.02543e-07 (mkt_cap)
- 0.00951100(pe_rat)

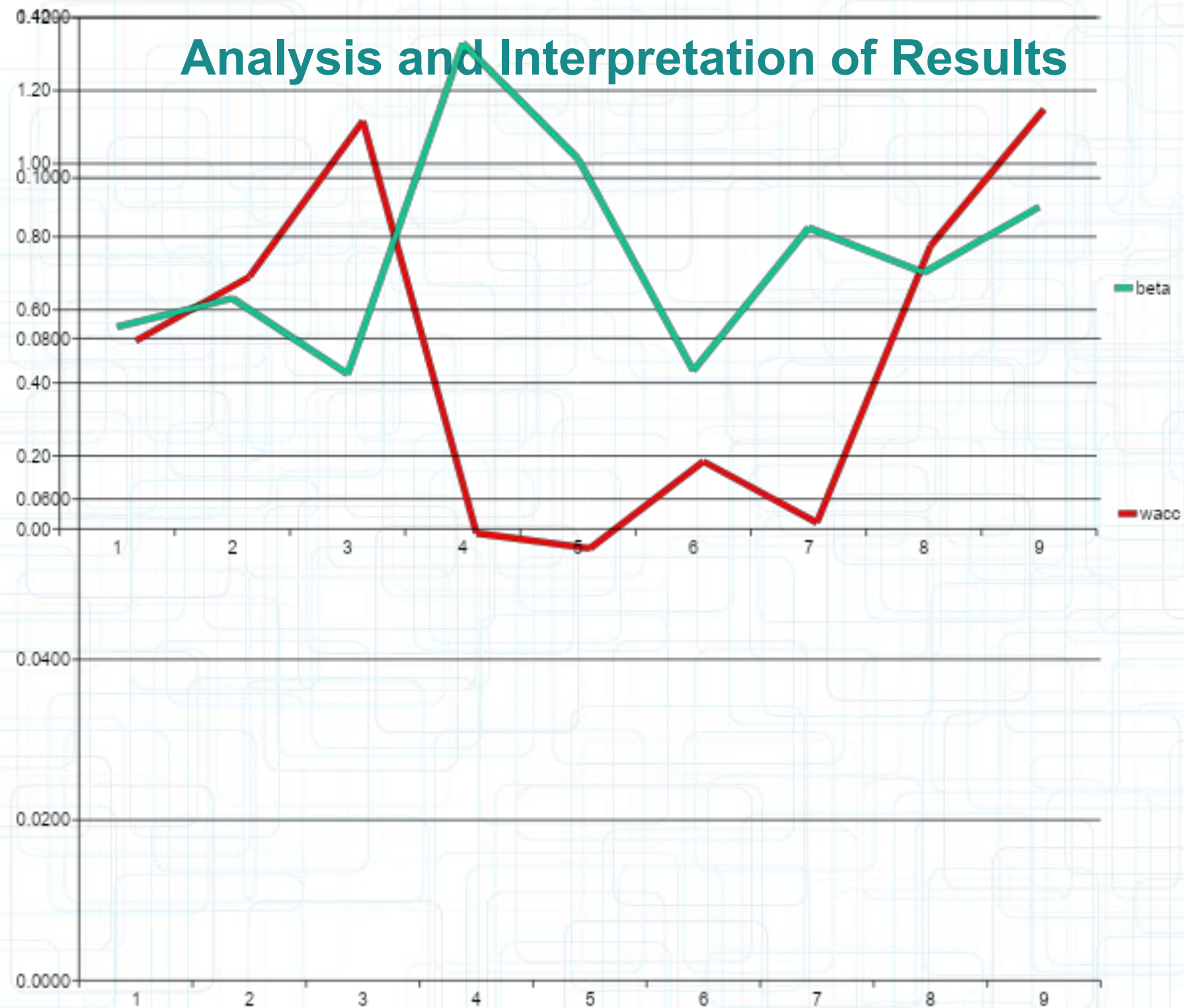
Analysis and Interpretation of Results



Analysis and Interpretation of Results



Analysis and Interpretation of Results



Analysis and Interpretation of Results

Test for Heteroskedasticity and Multicollinearity

Heteroskedasticity

Test statistic: $TR^2 = 49.228788$,

with p-value = $P(\text{Chi-square}(34) > 49.228788) = 0.044136$

Multicollinearity

l_revenue 1.737

fin_lev 13.385

fin_lev__2 11.702

asset_turn 1.212

pe_rat 1.024

mkt_cap 1.559

beta 1.105

Analysis and Interpretation of Results

CORRELATION			null
fin lev	-	pe rat	accept
fin lev	-	mkt cap	accept
fin lev	-	wacc	accept
REGRESSION			null
wacc	-	fin lev	reject
wacc	-	mkt cap	accept
wacc	-	pe rat	accept
wacc	-	l_revenue	reject
wacc	-	beta	reject

Conclusion and Recommendation

All correlations were insignificant under the null hypothesis.

Highly levered firms would incur a low cost of capital with the exception of exceeding the optimal debt level.

Increase the number of financial variables analyzed

Try to apply to a more actively traded sectors listed in the PSE